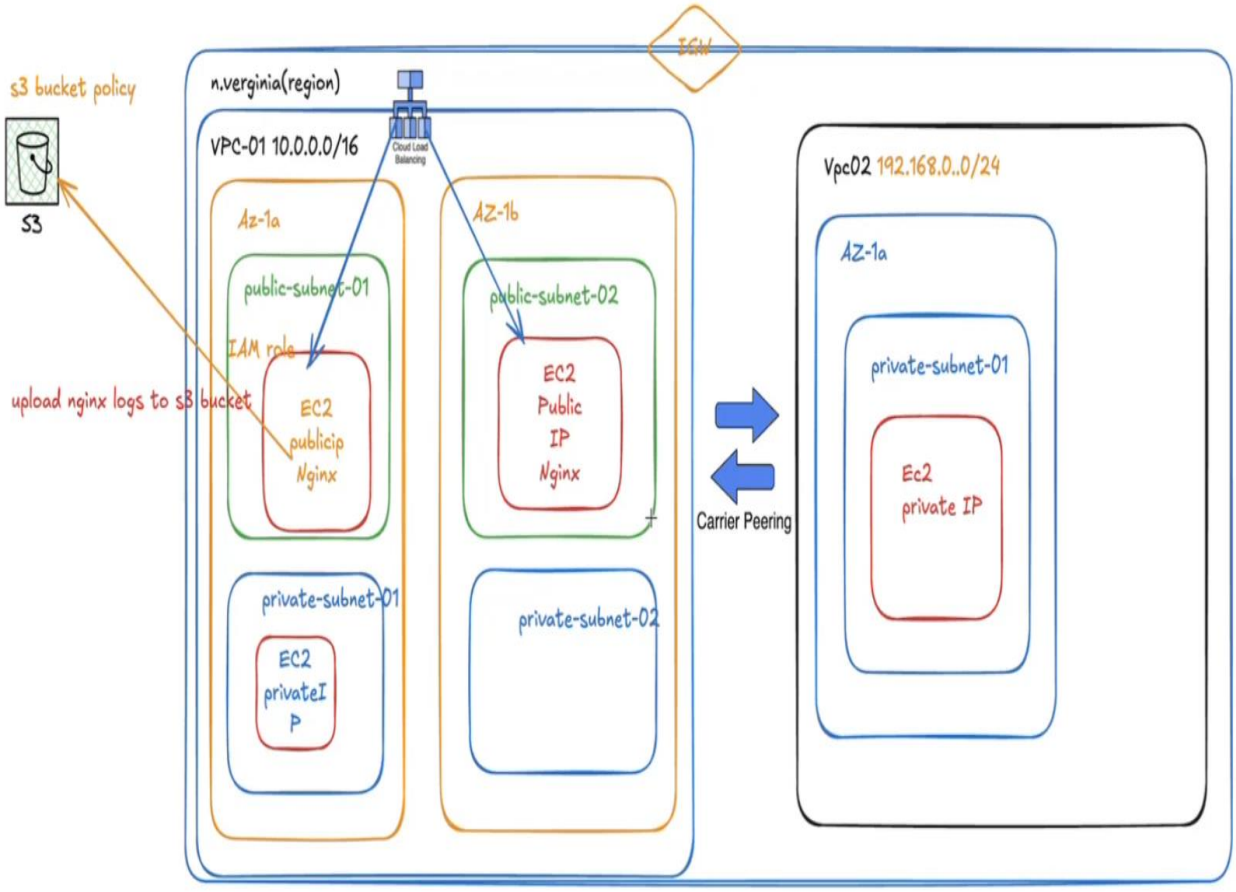
We Will Build This Infra



1- Create VPCs and its subnets

Your VPCs (2/3) Info

Last updated about 1 hour ago | Actions | Create VPC

Find VPCs by attribute or tag

Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set	Main route table
VPC-01-vpc	vpc-0d85a8645c602e35e	Available	Off	10.0.0.0/16	–	dopt-0086ed1fe455575...	rtb-027ed3b3493
–	vpc-0e2ef32e45607dec2	Available	Off	172.31.0.0/16	–	dopt-0086ed1fe455575...	rtb-0bfc1ac262c9
VPC-02-vpc	vpc-008232db8005e58de	Available	Off	192.168.0.0/24	–	dopt-0086ed1fe455575...	rtb-07a4d88540c

2- Create VPC Peering

Peering connections (1/1) Info

Actions | Create peering connection

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC	Accepter VPC	Requester CIDRs	Accepter CIDRs	Requester
my-peering	pcx-082ef5f92311cc1a1	Active	vpc-0d85a8645c602e35e / VPC...	vpc-008232db8005e58de / VP...	10.0.0.0/16	192.168.0.0/24	42060683

3- Create EC2s and install Nginx web server on only public EC2s using user data

Instances (4) Info

Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...
public-EC2-01	i-09b906d05d579e859	Stopping	t2.micro	–	View alarms +	us-east-1a	ec2-54-234-226-251.co...	54.234.226.251
private-EC2-01	i-066537a576858d931	Stopped	t2.micro	–	View alarms +	us-east-1a	–	–
private-EC2-02	i-05d00eeddbdbf29b7	Stopped	t2.micro	–	View alarms +	us-east-1a	–	–
public-EC2-02	i-057c150bc7a5eaa26	Stopping	t2.micro	–	View alarms +	us-east-1b	ec2-67-202-36-104.co...	67.202.36.104

4- Create Security Group for public EC2s

Inbound rules (1)

Manage tags | Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
–	sgr-0e35c2195f5819180	IPv4	All traffic	All	All	0.0.0.0/0	–

5- Create Target Group (TG)

MY-TG

Details

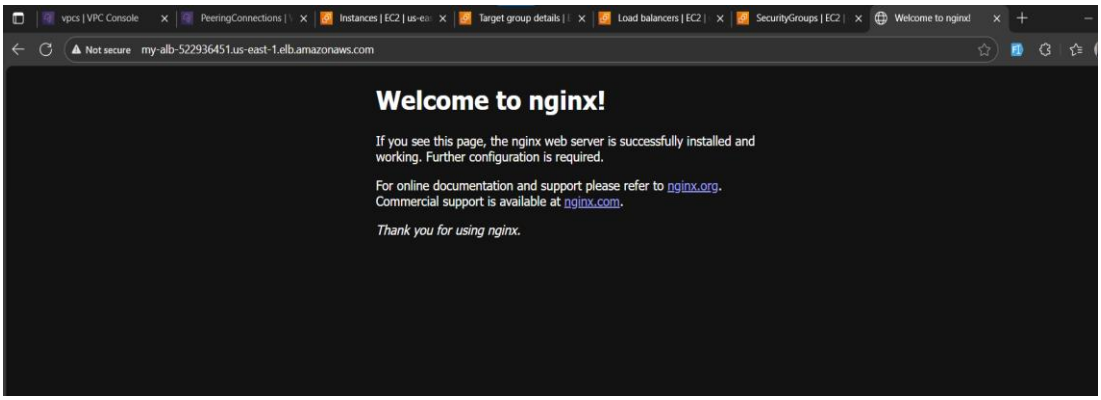
arn:aws:elasticloadbalancing:us-east-1:420606830171:targetgroup/MY-TG/080028d7793fff04

Target type	Protocol : Port	Protocol version	VPC
Instance	HTTP: 80	HTTP1	vpc-0d85a8645c602e35e
IP address type	**Load balancer**		
IPv4	MY-ALB		

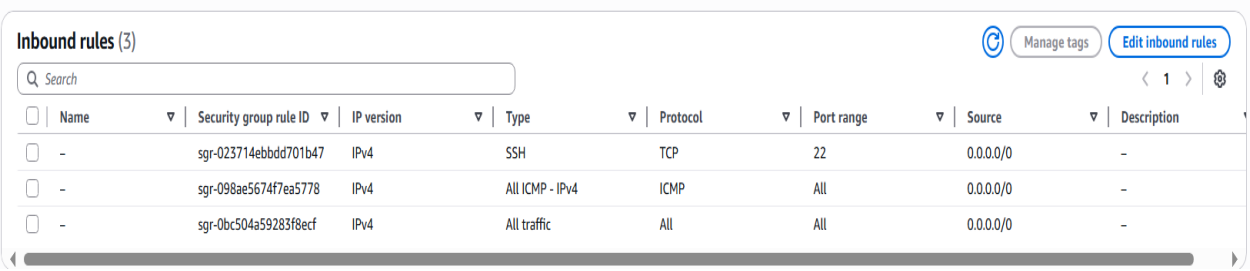
2	2	0	0	0	0
Total targets	Healthy	Unhealthy	Unused	Initial	Draining
	0 Anomalous				

6- Create Application Load Balancer (ALB)

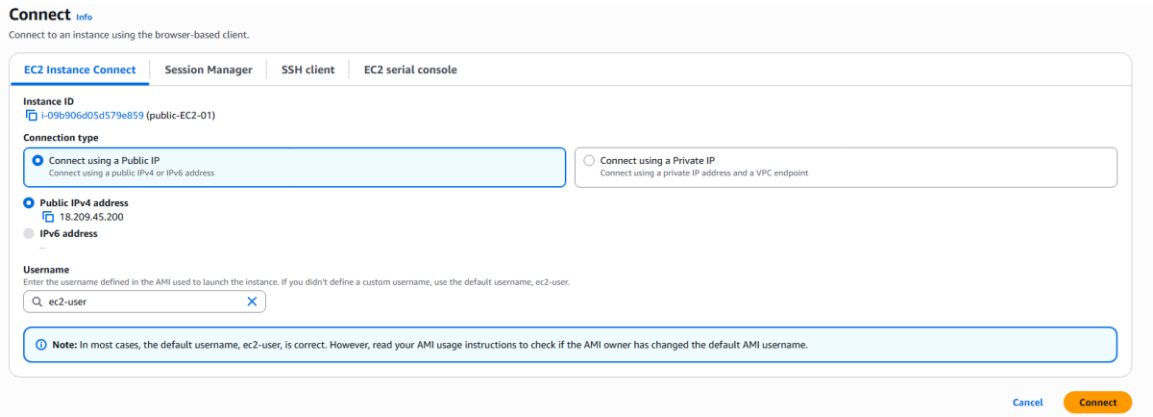
7- Hit the DNS of ALB in browser



8- Create Security Group for private EC2s



9- Connect to public EC2 in VPC-01



10- Copy the key of private EC2 in VPC-01 into public EC2 in VPC-01

```
2025-10-14  19:19.54  /home/mobaxterm/Desktop  scp -i EC2.pem ~/Desktop/private-EC2-01.pem ec2-user@54.157.215.183:/home/ec2-user/
private-EC2-01.pem                                                                                    100% 1678    9.7KB/s   00:00
```

11- Check the existence of key

```
[root@ip-10-0-2-55 ec2-user]# pwd
/home/ec2-user
[root@ip-10-0-2-55 ec2-user]# ls
private-EC2-01.pem
```

12- Connect from public EC2 in VPC-01 to private EC2 in VPC-01

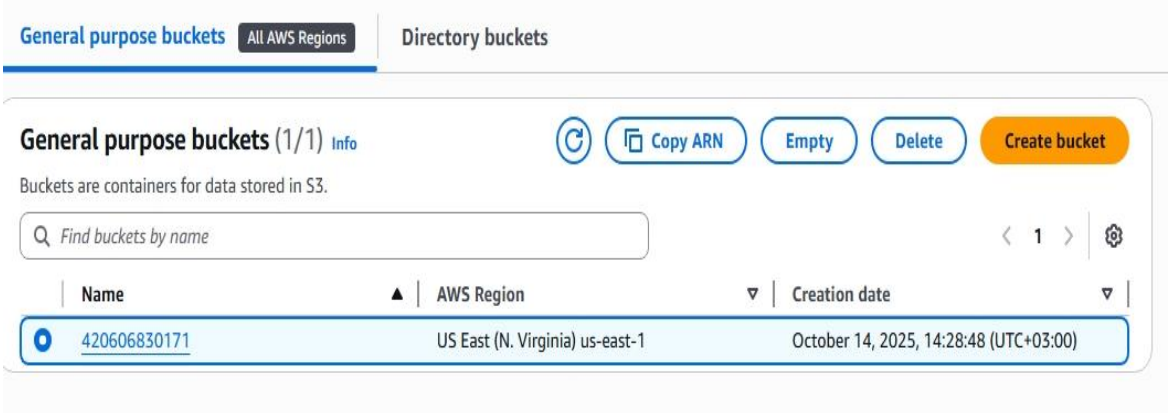
➔ **Now you are on the private EC2 in VPC-01**

```
[root@ip-10-0-2-55 ec2-user]# ssh -i private-EC2-01.pem ec2-user@10.0.132.87
The authenticity of host '10.0.132.87 (10.0.132.87)' can't be established.
ED25519 key fingerprint is SHA256:vgnf/8+c9nmzzWQYr2zqiAsgfqW4BNxcRaMm6Hhs7Ss.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '10.0.132.87' (ED25519) to the list of known hosts.
       ,     #_
       ~\_  ####_        Amazon Linux 2023
      ~~  \_#####\
      ~~     \###|
      ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
       ~~       V~' '->
        ~~~         /
          ~~._.   _/
             _/ _/
           _/m/'
```

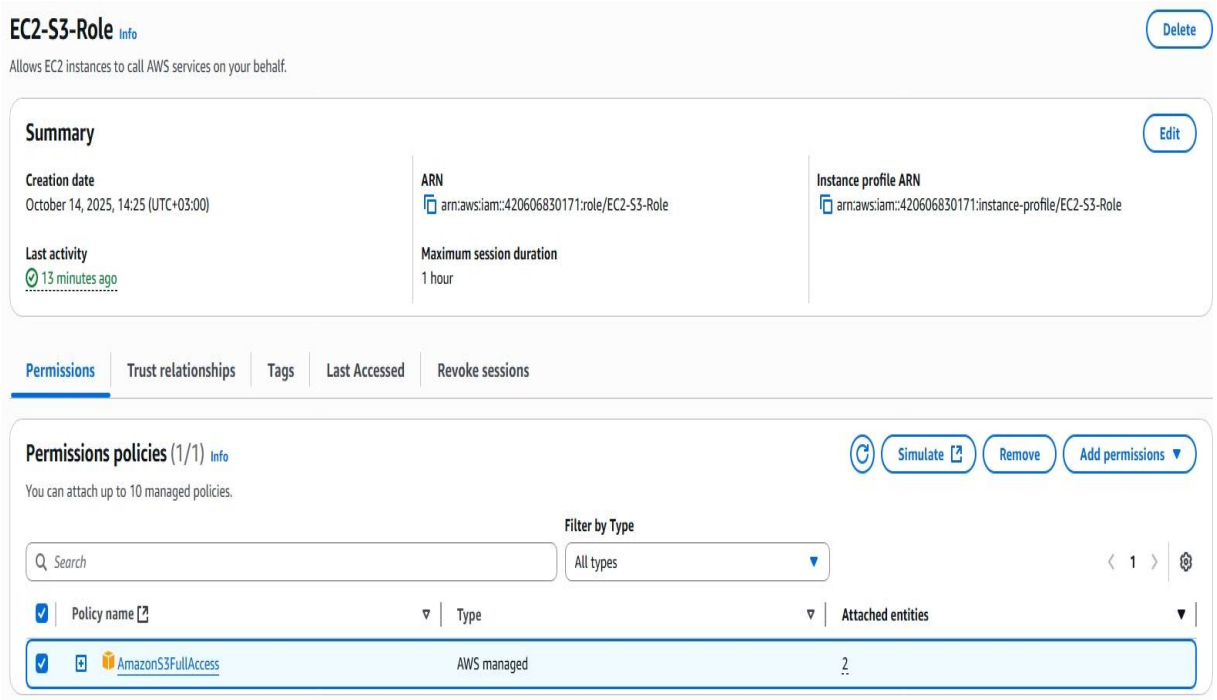
13- **Now you can ping from private EC2 in VPC-01 to private EC2 in VPC-02 (Peering)**

```
[ec2-user@ip-10-0-132-87 ~]$ ping 192.168.0.142
PING 192.168.0.142 (192.168.0.142) 56(84) bytes of data.
64 bytes from 192.168.0.142: icmp_seq=1 ttl=127 time=2.09 ms
64 bytes from 192.168.0.142: icmp_seq=2 ttl=127 time=1.44 ms
64 bytes from 192.168.0.142: icmp_seq=3 ttl=127 time=0.786 ms
64 bytes from 192.168.0.142: icmp_seq=4 ttl=127 time=1.06 ms
64 bytes from 192.168.0.142: icmp_seq=5 ttl=127 time=0.750 ms
64 bytes from 192.168.0.142: icmp_seq=6 ttl=127 time=1.07 ms
64 bytes from 192.168.0.142: icmp_seq=7 ttl=127 time=1.02 ms
64 bytes from 192.168.0.142: icmp_seq=8 ttl=127 time=0.722 ms
64 bytes from 192.168.0.142: icmp_seq=9 ttl=127 time=0.793 ms
```

14- Create S3 Bucket with account ID



15- Create IAM role for EC2



16- Create Bucket Security For S3 Bucket

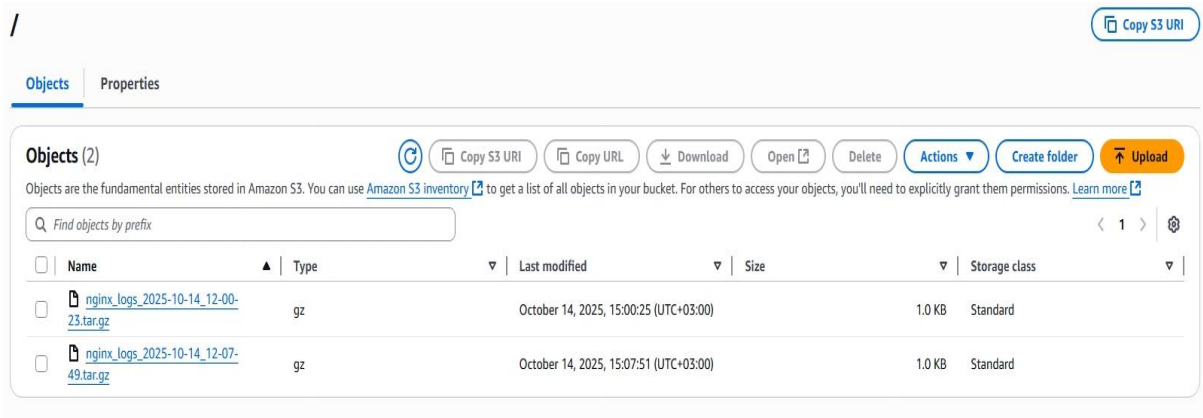
```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "AllowEC2AccessToS3Bucket",
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::420606830171:role/EC2-S3-Role"
            },
            "Action": [
                "s3:PutObject",
                "s3:GetObject",
                "s3:ListBucket"
            ],
            "Resource": [
                "arn:aws:s3:::420606830171",
                "arn:aws:s3:::420606830171/*"
            ]
        }
    ]
}
```

17- Create script to put Nginx logs on the S3 Bucket

Through this command -> nano /usr/local/bin/upload_nginx_logs.sh

```
#!/bin/bash

# Variables
BUCKET_NAME="420606830171"
INSTANCE_ID=$(curl -s http://169.254.169.254/latest/meta-data/instance-id)
LOG_DIR="/var/log/nginx"
DATE=$(date +"%Y-%m-%d_%H-%M-%S")

# Compress logs
tar -czf /tmp/nginx_logs_${DATE}.tar.gz -C $LOG_DIR .

# Upload file on S3
aws s3 cp "/tmp/nginx_logs_${DATE}.tar.gz" "s3://${BUCKET_NAME}/${INSTANCE_ID}/nginx_logs_${DATE}.tar.gz"

# Delete file form EC2 after uploading on S3
rm -f /tmp/nginx_logs_${DATE}.tar.gz
```

18- Logs Files will be created here -> 420606830171/



19- Log File like this

```
nginx_logs_2025-10-14_12-07-49 - Notepad
File  Edit  Format  View  Help
./                                                                                                  0000711 0000000 0000000 00000000000
15073225222 007701  5                                                                                                    ustar   root
                    root
                                                                                                ./error.log
                                                                    0000644 0000000 0000000 00000003462 15073431544 011560  0
                                                                                    ustar   root                            root

                                                                2025/10/13 16:36:34 [notice] 3898#3898: using the "epoll" event method
2025/10/13 16:36:34 [notice] 3898#3898: nginx/1.28.0
2025/10/13 16:36:34 [notice] 3898#3898: OS: Linux 6.1.153-175.280.amzn2023.x86_64
2025/10/13 16:36:34 [notice] 3898#3898: getrlimit(RLIMIT_NOFILE): 65535:65535
2025/10/13 16:36:34 [notice] 3947#3947: start worker processes
2025/10/13 16:36:34 [notice] 3947#3947: start worker process 3952
2025/10/13 16:40:01 [error] 3952#3952: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client:
197.43.206.103, server: _, request: "GET /favicon.ico HTTP/1.1", host: "98.89.21.112", referrer: "http://98.89.21.112/"
2025/10/13 16:47:17 [error] 3952#3952: *10 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client:
10.0.1.126, server: _, request: "GET /favicon.ico HTTP/1.1", host: "my-alb-522936451.us-east-1.elb.amazonaws.com", referrer: "http://my-
alb-522936451.us-east-1.elb.amazonaws.com/"
2025/10/13 16:52:33 [notice] 3947#3947: signal 3 (SIGQUIT) received from 1, shutting down
2025/10/13 16:52:33 [notice] 3952#3952: gracefully shutting down
2025/10/13 16:52:33 [notice] 3952#3952: exiting
2025/10/13 16:52:33 [notice] 3952#3952: exit
2025/10/13 16:52:33 [notice] 3947#3947: signal 17 (SIGCHLD) received from 3952
2025/10/13 16:52:33 [notice] 3947#3947: worker process 3952 exited with code 0
2025/10/13 16:52:33 [notice] 3947#3947: exit
2025/10/14 11:26:28 [notice] 2153#2153: using the "epoll" event method
2025/10/14 11:26:28 [notice] 2153#2153: nginx/1.28.0
2025/10/14 11:26:28 [notice] 2153#2153: OS: Linux 6.1.153-175.280.amzn2023.x86_64
2025/10/14 11:26:28 [notice] 2153#2153: getrlimit(RLIMIT_NOFILE): 65535:65535
2025/10/14 11:26:28 [notice] 2159#2159: start worker processes
2025/10/14 11:26:28 [notice] 2159#2159: start worker process 2160

                                                                ./access.log
                                                    0000644 0000000 0000000 00000007473 15073436137 011701  0
                                                                    ustar   root                            root

                                        197.43.206.103 - - [13/Oct/2025:16:40:01 +0000] "GET / HTTP/1.1" 200 615 "-" "Mozilla/5.0 (Windows NT 10.0; Win64;
x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0" "-"
197.43.206.103 - - [13/Oct/2025:16:40:01 +0000] "GET /favicon.ico HTTP/1.1" 404 3650 "http://98.89.21.112/" "Mozilla/5.0 (Windows NT 10.0;
Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/141.0.0.0 Safari/537.36 Edg/141.0.0.0" "-"
```

20- Cronjob to save Nginx logs in this path automatically every 5 minutes

```
[root@ip-10-0-2-55 ec2-user]# sudo crontab -l
*/5 * * * * /usr/local/bin/upload_nginx_logs.sh >> /var/log/upload_logs.log 2>&1
```